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AXIAL FAN

Abstract

An axial fan includes a fan frame, a motor and a fan wheel. The fan frame comprises an outer frame body and a mounting portion, wherein the outer frame body and the mounting portion are connected by a plurality of ribs. The motor is disposed at the mounting portion and has at least a cable across the outer frame body and the mounting portion, wherein at least a first rib of the ribs has an airflow mitigation structure. The motor is connected to the fan wheel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application Ser. No. 63/559,882, filed Mar. 1, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

Field of Disclosure

[0002] The present disclosure relates to an axial fan, and more particularly to an axial fan with a noise reduction functionality.

Description of Related Art

[0003] The ribs of the fan frame of a conventional axial fan are located on an air outlet side of the fan frame, and there is no rib structure on the air inlet side of the fan frame. If it is necessary to prevent the rotating impeller from being interfered with on the air inlet side, a fan guard needs to be installed over there. However, adding a fan guard will create additional noise.

SUMMARY

[0004] The present disclosure provides an axial fan to deal with the needs of the prior art problems.

[0005] In one or more embodiments, an axial fan includes a fan frame, a motor, and an impeller.

The fan frame includes an outer frame body and a mounting portion, wherein the outer frame body and the mounting portion are connected by a plurality of ribs. The motor is located on the mounting portion and has an electrical connection cable spanning between the outer frame body and the mounting portion, wherein a first rib of the ribs closest to the electrical connection cable has an arc-shaped airflow mitigation structure. The impeller is pivotally connected to the motor.

[0006] In one or more embodiments, the impeller is driven by the motor to rotate such that airflows are input from an air inlet side of the fan frame and output from an air outlet side of the fan frame, the airflow mitigation structure is located at the air inlet side of the first rib.

[0007] In one or more embodiments, the ribs, except for the first rib, comprises a plurality of second ribs having airflow guiding inclined surfaces.

[0008] In one or more embodiments, the airflow guiding inclined surface of the first rib is opposite to a rotation direction of the impeller of the axial fan.

[0009] In one or more embodiments, the electrical connection cable comprises a flexible printed circuit board.

[0010] In one or more embodiments, an axial flow channel is defined among the outer frame body, the mounting portion and the ribs. The outer frame body has a chamfered peripheral edge at an inner edge of the air outlet side facing the axial flow channel.

[0011] In one or more embodiments, the ribs include three ribs, and three central angles, formed by lines connecting the three ribs and a center of the mounting portion, are not equal.

[0012] In one or more embodiments, the impeller includes a plurality of blades, and gap widths between free ends of the blades have three different sizes.

[0013] In one or more embodiments, a windward side of each blade has a plurality of convex flow-guiding structures.

[0014] In one or more embodiments, an axial fan includes a fan frame, a motor, and an impeller.

The fan frame includes an outer frame body and a mounting portion, wherein the outer frame body and the mounting portion are connected by a plurality of ribs. The motor is located on the mounting portion and has an electrical connection cable overlapping a first rib of the ribs. The impeller is pivotally connected to the motor.

[0015] In one or more embodiments, the first rib has a greater width than that of the other ribs of the ribs.

[0016] In one or more embodiments, the first rib has airflow guiding inclined surfaces.

[0017] In one or more embodiments, the first rib has a greater width than that of the electrical connection cable.

[0018] In one or more embodiments, the impeller includes a plurality of blades, and a windward side of each blade has a plurality of convex flow-guiding structures.

[0019] In sum, the axial fan disclosed herein has the ribs of the fan frame designed to be on the air inlet side so as to avoid using the fan guard, and the ribs of the fan frame are designed to have a noise reduction function. In addition, the gap widths between the free ends of the blades of the axial fan have different sizes and have a plurality of convex flow-guiding structures on the windward side, which also has a noise reduction function.

[0020] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0022] FIG. 1 illustrates a perspective view of an axial fan according to an embodiment of the present disclosure;

[0023] FIG. 2 illustrates a perspective view of the axial fan in FIG. 1 with its impeller removed;

[0024] FIG. 3 illustrates a fan frame of the axial fan in FIG. 1;

[0025] FIG. 4 illustrates a side view of a 7-blade impeller of an axial fan according to some embodiments of the present disclosure;

[0026] FIG. 5 illustrates three cross-sectional views of the 7-blade impeller in FIG. 4 along three section lines;

[0027] FIG. 6 illustrates a side view of a 9-blade impeller of an axial fan according to some embodiments of the present disclosure;

[0028] FIG. 7 illustrates three cross-sectional views of the 9-blade impeller in FIG. 6 along three section lines;

[0029] FIG. 8 illustrates a top view of the 9-blade impeller in FIG. 6; and

[0030] FIG. 9 illustrates a perspective view of an axial fan with its impeller removed according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0031] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts. In the embodiments and patent claims, unless otherwise specified in the context, “a”, “an” and “the” may refer to a single or plural number.

[0032] Reference is made to FIG. 1, which illustrates a perspective view of an axial fan **100** according to an embodiment of the present disclosure. The axial fan **100** includes a fan frame **110**, a motor **150** (see FIG. 2) and an impeller **120**. The impeller **120** is driven by the motor **150** to rotate so that airflows are input from the air inlet side of the fan frame **110** and output from the air outlet side. As shown in the figure, the arrow A shows the direction of airflows in and out of impeller **120**. The fan frame **110** includes an outer frame body **110a** and a mounting portion **110b**, wherein the outer frame body **110a** and the mounting portion **110b** are connected by a plurality of ribs **112**. In some embodiments of the present disclosure, the ribs **112** include a rib **112a**, a rib **112b**, and a rib **112c**. In some embodiments of the present disclosure, the rib **112a** has an airflow guiding inclined surface **116a**, and the rib **112b** has an airflow guiding inclined surface **116b**. Taking the rib **112a** as

an example, the airflow guiding inclined surface **116a** thereof presents an inclined surface in the direction of the air inlet side. The inclined surface generally faces a direction **r**, which is opposite to a rotation direction **R** of the impeller **120**. This design helps reduce the turbulence generated by the axial fan during operation and further improves high-frequency noise.

[0033] Referring again to FIG. 1, the rib **112c** is closest to the electrical connection cable **114** and has an arc-shaped airflow mitigation structure **113**. The airflow mitigation structure **113** is located at the air inlet side of the rib **112c**, and can also be used to reduce the high-frequency noise generated when the axial fan is operating. In addition, the outer frame body **110a** may further include a connection cable buckle **117** for receiving the electrical connection cable **114**.

[0034] Reference is made to FIG. 2, which illustrates a perspective view of the axial fan **100** in FIG. 1 with its impeller **120** removed. This figure is a perspective view from the air outlet side. Referring FIGS. 1 and 2, the motor **150** is located on the circuit board **152** and is located on the mounting portion **110b**, and the electrical connection cable **114** extending from the motor **150** spans between the outer frame body **110a** and the mounting portion **110b**. The motor **150** is pivotally connected to the impeller **120**. In some embodiments of the present disclosure, the electrical connection cable **114** is a flexible printed circuit board. The electrical connection cable **114** not only spans the outer frame body **110a** and the mounting portion **110b**, but also has an electrical connector head **156** for connecting to an external control source. The outer frame body **110a** has a chamfered peripheral edge **111a** at an inner edge **111b** of the air outlet side facing the axial flow channel **115**, which helps to increase the air flow rate of the axial flow channel and reduce the noise generated when the axial fan is operating.

[0035] Reference is made to FIG. 3, which illustrates a fan frame **110** of the axial fan in FIG. 1. This figure is a view from the air outlet side of the fan frame **110**. In some embodiments of the present disclosure, each of the ribs **112a**, **112b**, **112c** has arc-shaped chamfers **A1** and **A2** on both sides of the connection between the ribs **112a**, **112b**, **112c** and the outer frame body **110a**. Since the viewing angle width of the arc-shaped chamfers **A1** and **A2** is larger, it is beneficial for the structural strength of the connection between the ribs and the outer frame body to enhance the shock resistance when falling, and further reduces the high-frequency noise generated by the operation of the axial fan.

[0036] Continuing with reference to FIG. 3, in some embodiments of the present disclosure, an axial flow channel **115** is formed between the outer frame body **110a**, the mounting portion **110b**, and the ribs **112a**, **112b**, and **112c**. The three central angles **A3**, **A4**, and **A5** formed by the lines connecting the ribs **112a**, **112b**, and **112c** and a center of the mounting portion **110b** are not equal, or the central angle **A3** is greater than the central angle **A4**, and the central angle **A4** is greater than the central angle **A5**. The change in the size of the axial flow channel **115** results in different air inlet velocities, which helps reduce noise by weakening the mutual influence of broadband noise at different wind speeds.

[0037] Reference is made to FIG. 4, which illustrates a side view of a 7-blade impeller **120a** of an axial fan according to some embodiments of the present disclosure. The impeller **120a** includes a plurality of blades **121** with a vertical height **H1**, and section lines (A-A, B-B, C-C) divide the vertical height **H1** of the impeller **120a** into four equal parts. The windward side **121a** of each blade **121** includes a plurality of convex flow-guiding structures, but the leeward side **121b** of each blade **121** is substantially smooth without any convex flow-guiding structures. The plurality of convex flow-guiding structures on the windward side **121a** of the blade **121** are used to reduce the high-frequency noise generated when the axial fan is operating.

[0038] Reference is made to FIG. 5, which illustrates three cross-sectional views of the 7-blade impeller in FIG. 4 along three section lines (A-A, B-B, C-C). Referring to both FIGS. 4 and 5, in the cross-sectional view taken along section line A-A, a plurality of convex flow-guiding structures (**122a**, **122b**) are shown on the windward side **121a**. In the cross-sectional view along the section line B-B, a plurality of convex flow-guiding structures (**122a**, **122b**) are still shown on the

windward side **121a**. In the cross-sectional view along the section line C-C, the windward side **121a** does not have the convex flow-guiding structure (**122a**, **122b**). In summary, the convex flow-guiding structure (**122a**, **122b**) of the impeller **120a** are only distributed on the windward side **121a** of the section lines A-A and B-B, that is, they are only distributed at three quarters of the vertical height H1 from the windward surface.

[0039] Reference is made to FIG. 6, which illustrates a side view of a 9-blade impeller **120b** of an axial fan according to some embodiments of the present disclosure. The impeller **120b** includes a plurality of blades **123**, forming a vertical height H2, and the section lines (A-A, B-B, C-C) divide the vertical height H2 of the impeller **120a** into four equal parts. The windward side **123a** of each blade **123** includes a plurality of protruding flow-guiding structures, but the leeward side **123b** of each blade **123** is substantially smooth. The plurality of convex flow-guiding structures on the windward side **123a** of the blade **123** are used to reduce the high-frequency noise generated when the axial fan is operating.

[0040] Reference is made to FIG. 7, which illustrates three cross-sectional views of the 9-blade impeller **120b** in FIG. 6 along three section lines (A-A, B-B, C-C). Referring to both FIGS. 6 and 7, in the cross-sectional view along the section line A-A, a plurality of convex flow-guiding structures (**124a**, **124b**) are shown on the windward side **123a**, and the leeward side **123b** is smooth. In the cross-sectional view along the section line B-B, a plurality of convex flow-guiding structures (**124a**, **124b**) are still shown on the windward side **123a**, while the leeward side **123b** is smooth. In the cross-sectional view along the section line C-C, a plurality of convex flow-guiding structures (**124a**, **124b**) are still shown on the windward side **123a**, while the leeward side **123b** is smooth. Based on the three cross-sectional views, the windward side **123a** of each blade **123** has at least a plurality of convex flow-guiding structures (**124a**, **124b**).

[0041] Referring again to FIG. 7, the convex flow-guiding structures (**124a**, **124b**) on the windward side **123a** of the impeller **120b** are distributed on the windward side **123a**, and at a vertical height H2, the convex flow-guiding structures (**124a**, **124b**) have different heights. Furthermore, the convex flow-guiding structures (**124a**, **124b**) have a maximum height in the cross-sectional view along the cross-sectional line B-B, and have a smaller height in the cross-sectional view along the cross-sectional line A-A and in the cross-sectional view along the cross-sectional line C-C. In other words, the height of the convex flow-guiding structure gradually increases from the cross section of the section line A-A to the cross section of the section line B-B, which reaches the maximum, and then gradually decreases to the cross section of the section line C-C.

[0042] Reference is made to FIG. 8, which illustrates a top view of the 9-blade impeller **120b** in FIG. 6. In some embodiments of the present disclosure, the impeller **120b** includes a plurality of blades **123**, and the gap widths (D1, D2, D3) between free ends of adjacent blades of the blades **123** have three different sizes. The design of different gap widths (D1, D2, D3) between adjacent blades **123** is also helpful in reducing the high-frequency noise generated when the axial fan is operating.

[0043] Reference is made to FIG. 9, which illustrates a perspective view of an axial fan with its impeller removed according to another embodiment of the present disclosure. The difference between this example and the embodiments of FIGS. 1 and 2 lies in the design of the fan frame **110'**. Specifically, the specific structure of the ribs of the fan frame **110'** is different. The ribs (**112a**, **112b**) of fan frame **110'** and fan frame **110** are the same and are both designed to reduce operating noise, but the ribs **112d** and **112c** are different. The rib **112a** has an airflow mitigation structure **113** to reduce the high frequency noise generated when the axial fan is operating. The rib **112d** is designed to overlap the electrical connector **152a**. The circuit board **152** is electrically coupled to the circuit board **154** via the electrical connector **152a** and is connected to an external control source via the electrical connector head **156**. In some embodiments of the present disclosure, the rib **112d** has a larger width W2 than the other ribs (**112b**, **112c**). In some embodiments of the present disclosure, the width W2 of the rib **112d** is greater than the width W1 of the electrical

connector **152a**. In some embodiments of the present disclosure, the electrical connector **152a** may be molded into the rib **112d** by an injection molding process. The motor **150** is located on the circuit board **152** and is located on the mounting portion **110b**. In above embodiments, the impeller (**120**, **120a**, **120b**) is pivotally connected to the motor **150**.

[0044] In sum, the axial fan disclosed herein has the ribs of the fan frame designed to be on the air inlet side so as to avoid using the fan guard, and the ribs of the fan frame are designed to have a noise reduction function. In addition, the gap widths between the free ends of the blades of the axial fan have different sizes and have a plurality of convex flow-guiding structures on the windward side, which also has a noise reduction function. Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

Claims

1. An axial fan comprising: a fan frame comprising an outer frame body and a mounting portion, wherein the outer frame body and the mounting portion are connected by a plurality of ribs; a motor disposed on the mounting portion, and having an electrical connection cable spanning between the outer frame body and the mounting portion, wherein a first rib of the ribs closest to the electrical connection cable has an airflow mitigation structure; and an impeller pivotally connected to the motor.
2. The axial fan of claim 1, wherein the impeller is driven by the motor to rotate such that airflows are input from an air inlet side of the fan frame and output from an air outlet side of the fan frame, the airflow mitigation structure is located at the air inlet side of the first rib.
3. The axial fan of claim 1, wherein the airflow mitigation structure is located at a side of the first rib closer to the electrical connection cable.
4. The axial fan of claim 1, wherein two sides of a connection between each of the ribs and the outer frame body have arc-shaped chamfers.
5. The axial fan of claim 1, wherein the ribs, except for the first rib, comprises a plurality of second ribs having airflow guiding inclined surfaces.
6. The axial fan of claim 1, wherein the electrical connection cable comprises a flexible printed circuit board.
7. The axial fan of claim 2, wherein an axial flow channel is defined among the outer frame body, the mounting portion and the ribs, the outer frame body has a chamfered peripheral edge at an inner edge of the air outlet side facing the axial flow channel.
8. The axial fan of claim 1, wherein the ribs include three ribs, and three central angles, formed by lines connecting the three ribs and a center of the mounting portion, are not equal.
9. The axial fan of claim 1, wherein the impeller includes a plurality of blades, and gap widths between free ends of the blades have three different sizes.
10. The axial fan of claim 9, wherein a windward side of each blade has a plurality of convex flow-guiding structures.
11. An axial fan comprising: a fan frame comprising an outer frame body and a mounting portion, wherein the outer frame body and the mounting portion are connected by a plurality of ribs; a motor disposed on the mounting portion, and having an electrical connection cable overlapping a first rib of the ribs; and an impeller pivotally connected to the motor.
12. The axial fan of claim 11, wherein the first rib has a greater width than that of the other ribs of the ribs.
13. The axial fan of claim 11, wherein the ribs, except for the first rib, comprises a plurality of second ribs having airflow guiding inclined surfaces.
14. The axial fan of claim 11, wherein the first rib has a greater width than that of the electrical

connection cable.

15. The axial fan of claim 11, wherein the impeller includes a plurality of blades, and a windward side of each blade has a plurality of convex flow-guiding structures.
